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Sports Data Soccer API - Remote Match Tracking EPTS Feed (MA49)

Get tracking data giving the coordinates of the players, and ball position during a given match.

Overview

The Soccer **Match Tracking feed** provides X/Y coordinate data showing the position of all players and ball on the pitch at a given point in time. The data is produced by StatsPerform's Opta Vision Imputation model based on broadcast video and is provided after the match has ended (post match) in **FIFA Electronic Performance and Tracking Systems (EPTS)** format.



Format Specification and Metadata Description - Opta Vision Match Tracking EPTS Feed (MA49) v2.2.0.pdf

The format consists of 2 files:

- Format specification - XML file containing metadata and data format specification,
- Raw data file - TXT file containing raw tracking data in a format that can be parsed line by line.

The files are explained below in separate sections. The feed offers URLs (links) to these files. You must copy and paste the URL into a web browser (into its address bar) to initiate the download.

XML:

```
<remoteMatchTrackingEpts>
  <matchInfo id="7ijuqohwgmp1bxdj1625sxwfe" ...>
    ... (sample is intentionally truncated) ...
  </matchInfo>
  <trackingFiles>
    <fps25>
      <metadata>http://api.stage.performfeeds.com/soccerdata/remotematchtrackingepts/download/remote/25fps/metadata_sd2s8nz1rnkb843p6hd7dbf8.xml?token=st=1709199834~exp=1709200434~acl=/*~hmac=f90db0cadaffab10a17014d882e2905509c5df4929959f892310a83daa2c4172</metadata>
    <trackingData>http://api.stage.performfeeds.com/soccerdata/remotematchtrackingepts/download/remote/25fps/sd2s8nz1rnkb843p6hd7dbf8.txt?token=st=1709199834~exp=1709200434~acl=/*~hmac=f90db0cadaffab10a17014d882e2905509c5df4929959f892310a83daa2c4172</trackingData>
    </fps25>
  </trackingFiles>
</remoteMatchTrackingEpts>
```

Required Compliance:

To comply with our API security and best practice logic for your API calls please refer to our **FAQs**. We also recommend that you take a look at our **API Overview and Authentication Guide**. You will learn the basics of how our API works and key concepts to get started.

How Often Should I Poll The Feed?

This is a post-match feed, so the data is provided in downloadable files; therefore, you should only call the feed only as often as you need to. You will be provided with a URL. You must copy and paste the URL from the feed output/response into a web browser (into its address bar) to initiate

POLLING FREQUENCY
Post-match

Note: This feed will launch with limited availability. Please ask your Account Manager for information about competition availability and about commercial legalities around this feed.

Remote Match Tracking EPTS (MA49) Guide

USAGE	FEED
GET match tracking data showing the pitch coordinates of the players, referee and ball during the given match	/soccerdata/remotematchtrackingepts/{ {fixtureUuid}}{queryParameters}

If your request URL is as follows - that is, it does not contain a fixture UUID - the feed returns an error:
/soccerdata/remotematchtrackingepts/{outletAuthKey}?_fmt={xml|json}&_rt={b|c}

Query Parameters And Example Request Parameters

This feed also makes use of the [Global query parameters](#), and you can combine them with the following query parameters (some of which can be used in conjunction with each other). However, you must query the `/remotematchtrackingepts` feed by fixture UUID.

The ability to query by a legacy Opta ID is not provided in the Match Tracking feed.

How to use query parameters and make successful requests:
In our guides, we use HTTPS syntax for our example GET requests (rather than cURL, for example). All URLs are case-sensitive - this means that URL strings must contain the correct case and operators. You MUST format your requests correctly and include certain information; otherwise, an error will be returned. You can find information about the API, including the base URL (`{protocol}://{requestDomain}/{feedResource}`), in the [main guide](#) and [API Overview and Authorisation guide](#).

- Include the following information in the URL or header and make sure it is correct for your outlet: your unique and valid `{outletAuthKey}` this must be after the Outlet Authentication Key (and endpoint, if applicable). for `_rt={mode}` (operating mode) and `_fmt={dataFormat}` (response format), entity query parameter and UUID.
- Begin the query parameter string with `?` (question mark) - this must be after the Outlet Authentication Key (and endpoint, if applicable). To build up a query string, make sure that each query parameter is separated by the `&` (ampersand) operator. Values must be separated by the correct operator, for example `&` (multiple values, 'AND'), or - if a parameter supports it - `,` (multiple values, 'OR') or `!` ('NOT').
- Use the required `_` (underscore) for any parameters listed with this prefix.
- Remove any placeholder value curly braces `{ }` from your calls (we include these as an example only).
- If using the JSONP format (`_fmt=jsonp`) you MUST use it in combination with the callback parameter (`_clbk`) and a fixed value (or where you use a different value in a different instance, you must do this as little as possible). Be aware that common JavaScript frameworks, such as jQuery, can default to use randomly-generated function names - you must make sure that these are overridden and define a fixed function name instead.

Query parameter	Value and description
<code>{fixtureUuid}</code> Fixture UUID	GET match tracking data by fixture UUID (path parameter method). Pass the fixture UUID in the request URL directly as the <code>{assetId}</code> (asset/resource ID). GET https://api.performfeeds.com/soccerdata/remotematchtrackingepts/{outletAuthKey}/{ {fixtureUuid} ?_rt={mode}&_fmt={dataFormat}

Sample Calls

Coming Soon

Feed Output

Here, you can find an explanation of all possible response fields and data. This is XML but fields in the feed response are similar for JSON.

▶<remoteMatchTrackingEpts> (XML only)

▶<matchInfo>

▶<description>

▶<sport>

▶<ruleset>

▶<competition>

▶<country>

▶<tournamentCalendar>

▶<stage>

▶<series>

▶<contestants> (XML only)

▶<contestant>

▶<country>

▶<venue>

▶<trackingFiles>

▶<fps25>

▶<metadata>

▶<trackingData>

Understanding The Tracking File Data

Each tracking file is a comma-separated value (CSV) file with a `.txt` file extension. The structure of the `.txt` files is in rows, then columns. When you download and open a file, you will see that it comprises many rows and columns of characters. For example, there could be 146,853 rows in the file.

For every request 2 links will be provided, one for meta data and one for the tracking file. Meta data provide information about what is inside the tracking file

Understanding The Format Specification File

Format specification file is a file with `.xml` extension. Its contents follow the [EPTS file format specification](#), but introduce a couple of new, custom elements and values.

Below is an example of format specification file, truncated for simplicity:

```
<?xml version="1.1" encoding="UTF-8"?>
<tns:FIFADataTransferFormatEPTS xmlns:tns="FIFADataTransferFormatEPTSNamespace/FIFADataTransferFormatEPTS" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="FIFADataTransferFormatEPTSNamespace/FIFADataTransferFormatEPTS FIFADataTransferFormatEPTS.xsd">
  <Metadata>
    <GlobalConfig>
      <FileDate>2025-09-22T08:27:35Z</FileDate>
      <ProviderName>StatsPerform</ProviderName>
      <FrameRate>25</FrameRate>
      <FileName>13ylro8dy443m52qxwxm9zis_epts_format_specification</FileName>
      <Version>2.2.0</Version>
      <TrackingType>Optical</TrackingType>
      <ProviderGlobalParameters>
        <ProviderParameter>
          <Name>match_uuid</Name>
          <Value>13ylro8dy443m52qxwxm9zis</Value>
        </ProviderParameter>
      </ProviderGlobalParameters>
    </GlobalConfig>
    <Sessions>
      <Session id="matchsession">
        <Start>2025-08-18T19:02:04Z</Start>
        <End>2025-08-18T20:57:12Z</End>
        <MatchParameters>
          <FieldSize>
            <Width>68.0</Width>
            <Length>105.0</Length>
          </FieldSize>
        </MatchParameters>
      </Session>
    </Sessions>
    <Teams>
      <Team id="48gk2hpqtsl6p9sx9kjhaydq4"/>
      <Team id="ehd2iemqmschhj2ec0vayztzz"/>
    </Teams>
    <DirectionsOfPlay>
      <Period id="1">
        <Team id="48gk2hpqtsl6p9sx9kjhaydq4">
          <DirectionOfPlay>LeftToRight</DirectionOfPlay>
        </Team>
        <Team id="ehd2iemqmschhj2ec0vayztzz">
          <DirectionOfPlay>RightToLeft</DirectionOfPlay>
        </Team>
      </Period>
      <Period id="2">
        <Team id="48gk2hpqtsl6p9sx9kjhaydq4">
          <DirectionOfPlay>RightToLeft</DirectionOfPlay>
        </Team>
        <Team id="ehd2iemqmschhj2ec0vayztzz">
          <DirectionOfPlay>LeftToRight</DirectionOfPlay>
        </Team>
      </Period>
    </DirectionsOfPlay>
    <Players>
      <Player id="18qxb54ol1s4mtgl7n3224ciy" teamId="ehd2iemqmschhj2ec0vayztzz">
        <ShirtNumber>24</ShirtNumber>
        <ProviderPlayerParameters>
          <ProviderParameter>
            <Name>type</Name>
            <Value>1</Value>
          </ProviderParameter>
        </ProviderPlayerParameters>
      </Player>
```

```

<Player id="2m5jzeksbet81tn3shk2exox6" teamId="ehd2iemqmschhj2ec0vayztzz">
  <ShirtNumber>17</ShirtNumber>
  <ProviderPlayerParameters>
    <ProviderParameter>
      <Name>type</Name>
      <Value>1</Value>
    </ProviderParameter>
  </ProviderPlayerParameters>
</Player>
<!-- other players definitions -->
</Players>
<FrameCountToVideoTimeMsMapping>
  <Frame frameCount="10000" videoTimeMs="646060"/>
  <Frame frameCount="10001" videoTimeMs="646094"/>
  <Frame frameCount="10002" videoTimeMs="646127"/>
  <Frame frameCount="10003" videoTimeMs=""/>
  <Frame frameCount="10004" videoTimeMs="646227"/>
  <!-- more frame mappings -->
  <Frame frameCount="1078904" videoTimeMs="7553791"/>
</FrameCountToVideoTimeMsMapping>
<Devices>
  <Device id="FieldLocator">
    <Name>OV CV Imputation</Name>
    <Sensors>
      <Sensor id="playerinfo">
        <Name>Info</Name>
        <Channels>
          <Channel id="uuid">
            <Name>player_uuid</Name>
          </Channel>
        </Channels>
      </Sensor>
      <Sensor id="position">
        <Name>Position</Name>
        <Channels>
          <Channel id="x">
            <Name>position_x, empty if none</Name>
          </Channel>
          <Channel id="y">
            <Name>position_y, empty if none</Name>
          </Channel>
          <Channel id="z">
            <Name>position_z, empty if none</Name>
          </Channel>
        </Channels>
      </Sensor>
      <Sensor id="cvinfo">
        <Name>CVInfo</Name>
        <Channels>
          <Channel id="visible">
            <Name>visible on screen</Name>
          </Channel>
          <Channel id="period">
            <Name>match period</Name>
          </Channel>
          <Channel id="optaevents">
            <Name>list of opta event ids separated by |, empty if none</Name>
          </Channel>
          <Channel id="teaminpossession">
            <Name>team in possession, empty if none</Name>
          </Channel>
          <Channel id="playerinpossession">
            <Name>player in possession, empty if none</Name>
          </Channel>
        </Channels>
      </Sensor>
    </Sensors>
  </Device>
</Devices>

```

```

    </Channel>
    <Channel id="videotimems">
      <Name>video time reference</Name>
    </Channel>
    <Channel id="speed">
      <Name>player or ball speed</Name>
    </Channel>
  </Channels>
</Sensor>
</Sensors>
</Device>
</Devices>
<PlayerChannels>
  <PlayerChannel channelId="uuid" id="35jaxclh329duzefna708jk5x_uuid" playerId="35jaxclh329duzefna708jk5x"/>
  <PlayerChannel channelId="x" id="35jaxclh329duzefna708jk5x_x" playerId="35jaxclh329duzefna708jk5x"/>
  <PlayerChannel channelId="y" id="35jaxclh329duzefna708jk5x_y" playerId="35jaxclh329duzefna708jk5x"/>
  <PlayerChannel channelId="speed" id="35jaxclh329duzefna708jk5x_speed" playerId="35jaxclh329duzefna708jk5x"/>
  <PlayerChannel channelId="visible" id="35jaxclh329duzefna708jk5x_visible" playerId="35jaxclh329duzefna708jk5x"/>
  <PlayerChannel channelId="uuid" id="cnoomjxp6tna0mx2ycuptgixh_uuid" playerId="cnoomjxp6tna0mx2ycuptgixh"/>
  <PlayerChannel channelId="x" id="cnoomjxp6tna0mx2ycuptgixh_x" playerId="cnoomjxp6tna0mx2ycuptgixh"/>
  <PlayerChannel channelId="y" id="cnoomjxp6tna0mx2ycuptgixh_y" playerId="cnoomjxp6tna0mx2ycuptgixh"/>
  <PlayerChannel channelId="speed" id="cnoomjxp6tna0mx2ycuptgixh_speed" playerId="cnoomjxp6tna0mx2ycuptgixh"/>
  <PlayerChannel channelId="visible" id="cnoomjxp6tna0mx2ycuptgixh_visible" playerId="cnoomjxp6tna0mx2ycuptgixh"/>
  <!-- other players channels -->
</PlayerChannels>
</Metadata>
<DataFormatSpecifications>
  <DataFormatSpecification separator=":" startFrame="10000" endFrame="1032314">
    <StringRegister name="frameCount"/>
    <SplitRegister separator=";">
      <SplitRegister separator=",">
        <PlayerChannelRef playerId="35jaxclh329duzefna708jk5x_uuid"/>
        <PlayerChannelRef playerId="35jaxclh329duzefna708jk5x_x"/>
        <PlayerChannelRef playerId="35jaxclh329duzefna708jk5x_y"/>
        <PlayerChannelRef playerId="35jaxclh329duzefna708jk5x_speed"/>
        <PlayerChannelRef playerId="35jaxclh329duzefna708jk5x_visible"/>
      </SplitRegister>
      <SplitRegister separator=",">
        <PlayerChannelRef playerId="ceerfv28fygp7m2euv5n5uu39_uuid"/>
        <PlayerChannelRef playerId="ceerfv28fygp7m2euv5n5uu39_x"/>
        <PlayerChannelRef playerId="ceerfv28fygp7m2euv5n5uu39_y"/>
        <PlayerChannelRef playerId="ceerfv28fygp7m2euv5n5uu39_speed"/>
        <PlayerChannelRef playerId="ceerfv28fygp7m2euv5n5uu39_visible"/>
      </SplitRegister>
      <!-- other players data structures -->
    </SplitRegister>
    <SplitRegister separator=",">
      <BallChannelRef channelId="x"/>
      <BallChannelRef channelId="y"/>
      <BallChannelRef channelId="z"/>
      <BallChannelRef channelId="speed"/>
      <BallChannelRef channelId="visible"/>
      <BallChannelRef channelId="period"/>
      <BallChannelRef channelId="optaevents"/>
      <BallChannelRef channelId="teaminpossession"/>
      <BallChannelRef channelId="playerinpossession"/>
      <BallChannelRef channelId="videotimems"/>
    </SplitRegister>
  </DataFormatSpecification>
  <!-- other frame ranges data structure specifications -->
</DataFormatSpecifications>
</tns:FIFADataTransferFormatEPTS>

```

Metadata

Defines information necessary for understanding raw data file.

- **GlobalConfig**
 - **FileDate**: UTC timestamp when data was generated, for example `2025-09-22T08:27:35Z` (string)
 - **ProviderName**: provider name, always `StatsPerform` (string)
 - **FrameRate**: frame rate of raw data, always `25` (integer)
 - **FileName**: name of the format specification file, for example `13y1ro8dy443m52qwxrxm9zis_epts_format_specification` (string)
 - **Version**: version of the data format, for example `2.2.0` (string)
 - **TrackingType**: type of tracking technology used, always `Optical` (string)
 - **ProviderGlobalParameters**: custom global parameters added by StatsPerform
 - **ProviderParameter**: parameter containing match identifier
 - **Name**: parameter name, always `match_uuid` (string)
 - **Value**: SDAPI match UUID, for example `13y1ro8dy443m52qwxrxm9zis` (string)
- **Sessions**: specific phases within the event
 - **Session**: single session covering the full match
 - **[id]**: identifier of the session, always `matchsession` (string)
 - **Start**: UTC timestamp when the session started, for example `2025-08-18T19:02:04Z` (string)
 - **End**: UTC timestamp when the session ended, for example `2025-08-18T20:57:12Z` (string)
 - **MatchParameters**: parameters related to match
 - **FieldSize**: length and width of the field
 - **Length**: length of the field in meters, for example `105.0` (double)
 - **Width**: width of the field in meters, for example `68.0` (double)
- **Teams**: teams playing in the match
 - **Team**: single team
 - **[id]**: SDAPI UUID of the team, for example `48gk2hpqts16p9sx9kjhaydq4` (string)
- **DirectionsOfPlay**: direction of play for both teams and periods
 - **Period**: game period
 - **[id]**: 1 or 2 (string)
 - **Team**: team
 - **[id]**: team UUID, e.g. `48gk2hpqts16p9sx9kjhaydq4` (string)
 - **DirectionOfPlay**: LeftToRight or RightToLeft (string)
- **Players**: players appearing in the tracking data at any point in time
 - **Player**: single player
 - **[id]**: SDAPI UUID of the player, for example `18qxb54o11s4mtg17n3224ciy` (string)
 - **[teamId]**: identifier of the player's team as presented in metadata, for example, `ehd2iemqmschhj2ec0vayztzz` (string)
 - **ShirtNumber**: shirt (jersey) number of the player, for example `23` (string)
 - **ProviderPlayerParameters**: custom player parameters added by StatsPerform
 - **ProviderParameter**: parameter containing player type
 - **Name**: `type` (string)
 - **Value**: player type, one of `0` = home field player, `1` = away field player, `3` = home goalkeeper, `4` = away goalkeeper (string)
- **FrameCountToVideoTimeMsMapping**: mapping between `frame_count` and `video_time_ms` (new in metadata version 2.0.0)
 - **Frame**: single frame
 - **[frameCount]**: corresponding to `frame_count` from the data file
 - **[videoTimeMs]**: timestamp of the corresponding LiveCV frame or empty if no corresponding frame exists
- **Devices**: description of devices providing measurements presented in raw data file
 - **Device**: single virtual device with multiple sensors providing all measurements
 - **[id]**: device identifier, always `FieldLocator` (string)
 - **Name**: device name, always `OV CV Imputation` (string)
 - **Sensors**: sensors obtaining measurements
 - **Sensor**: sensor obtaining player information
 - **[id]**: sensor identifier, always `playerinfo` (string)
 - **Name**: sensor name, always `Info` (string)
 - **Channels**: measured attributes (channels)

- Channel:** SDAPI player UUID - **[id]:** channel identifier, always `uuid` (string), **Name:** channel name, always `player_uuid` (string)
 - Sensor:** sensor obtaining position (coordinates)
 - [id]:** sensor identifier, always `position` (string)
 - Name:** sensor name, always `Position` (string)
 - Channels:** measured attributes (channels)
 - Channel:** position X coordinate - **[id]:** channel identifier, always `x` (string), **Name:** channel name, always `position_x` (string)
 - Channel:** position Y coordinate - **[id]:** channel identifier, always `y` (string), **Name:** channel name, always `position_y` (string)
 - Channel:** position Z coordinate - **[id]:** channel identifier, always `z` (string), **Name:** channel name, always `position_z, -1 if none` (string)
 - Sensor:** sensor obtaining computer vision information
 - [id]:** sensor identifier, always `cvinfo` (string)
 - Name:** sensor name, always `CVInfo` (string)
 - Channels:** measured attributes (channels)
 - Channel:** marks if position is visible/detected from broadcast or determined by imputation model - **[id]:** channel identifier, always `visible` (string), **Name:** channel name, always `visible on screen` (string)
 - Channel:** numeric `match period` value - **[id]:** channel identifier, always `period` (string), **Name:** channel name, always `match period` (string)
 - Channel:** corresponding opta event reference - **[id]:** channel identifier, always `optaevents` (string) - **Name:** channel name, always `list of opta event ids separated by |, empty if none` (integer)
 - Channel:** team in possession - **[id]:** channel identifier, always `teaminpossession` (string) - **Name:** channel name, always `team in possession, empty if none` (string)
 - Channel:** player in possession - **[id]:** channel identifier, always `playerinpossession` (string) - **Name:** channel name, always `player in possession, empty if none` (string)
 - Channel:** match clock including milliseconds as unix timestamp units - **[id]:** channel identifier, always `videotimems` (string), **Name:** channel name, always `video time reference` (string)
 - Channel:** speed - **[id]:** channel identifier, always `speed` (string) - **Name:** channel name, always `player or ball speed` (string)
 - PlayerChannels:** association between attributes measured by devices and individual players
 - PlayerChannel:** single measured attribute - player association
 - [id]:** identifier of the association, for example, `9tc6mprxpu2rc5ptlp5ljw7h1_x` (string)
 - [channelId]:** identifier of the measured attribute (channel) as presented in sensor metadata, for example `x` (string)
 - [playerId]:** identifier of the player as presented in player metadata, for example, `9tc6mprxpu2rc5ptlp5ljw7h1` (string)

Data Format Specifications

Defines exact structure (parsing format) of each row in raw data file.

- DataFormatSpecification:** defines exact structure (parsing format) for frames ranging from `startFrame` to `endFrame` inclusive - different frame ranges can have different format (for example, during player substitutions)
 - [separator]:** top-level separator of data blocks, always `:` (string)
 - [startFrame]:** first frame the structure applies to, for example `10000` (integer)
 - [endFrame]:** last frame the structure applies to, for example `1032314` (integer)
 - StringRegister:** contains frame counter
 - [name]:** `frameCount` (string)
 - SplitRegister:** contains structure of combined data of all players in the frame
 - [separator]:** separator between data blocks of individual players, always `;` (string)
 - SplitRegister:** contains structure of data for a single player
 - [separator]:** separator between data parts of single player, always `,` (string)
 - PlayerChannelRef:** contains specific measured attribute for a specific player (`uuid`, `x`, `y`, `speed`, `visible`)
 - [playerChannelId]:** identifier of the single measured attribute - player association as presented in player channels metadata, for example `35jaxc1h329duzefna708jk5x_x` (string)
 - SplitRegister:** contains ball and frame-level data

- **[separator]**: separator between parts of ball and frame-level data, always `,` (string)
- **BallChannelRef**: contains specific measured ball/frame-level attribute (`x`, `y`, `z`, `speed`, `visible`, `period`, `optaevent`, `teaminpossession`, `playerinpossession`, `videotimems`)
 - **[channelId]**: identifier of the single measured attribute as presented in sensor metadata, for example `x` (string)

Understanding The Raw Data File

Raw data file is a file with a `.txt` extension. The structure of the `.txt` file is in rows, then columns. The file contains thousands of rows, each row contains tracking data for a single frame. There are 25 frames per second (fps).

The row is structured into 3 parts separated and finished by colon:

[Frame count]:[Player tracking data]:[Ball tracking and frame-level data]:

The individual parts are explained in separate sections below.

For example, one frame data:

```
12435:35jaxclh329duzefna708jk5x,9.8835,23.4084,1.3310,1;ceervf28fygp7m2euv5n5uu39,7.8188,18.8953,5.1791,1;6og6s61vrg51rfqk9ci
pylzp,15.3161,13.5224,4.7032,1;e8dina16u5kt5ho4u5igtmt,36.8300,4.7639,3.1484,0;f1ma05bi3t1hfbf1ylx5d9tyh,-9.6142,17.4655,4.0
098,0;dkwkmrmpb7rq4e660efcu5hat,14.1907,-13.1176,6.0789,1;dve7yj8usejn9x7pmdjvozje1,-41.3574,4.3697,2.0913,0;otan631zzfnq2sekt
kl3ywo9,-18.6350,6.4481,3.7403,0;2bkp1lcaxmvwoanxwo0775cd1,-8.2578,18.9799,4.9383,0;b275i2lg3anqae5lv9gmrzmqx,-18.3204,20.088
0,4.1192,0;dwumn5qew7ae53givnx2p892h,-9.2246,-5.6002,6.2606,0;15948izygnu32cu1pw6h5wp91,15.9913,-4.0549,5.7210,1;6ka58mn83rknx
80e4y2jk3949,13.1778,14.9963,5.8643,1;fij46jrf9fq43444gr1y8tl9l,-10.9865,28.6350,3.8852,0;28ucbvor6b0yxreioieehudfc9,-0.9159,7.8
798,5.5404,1;3dm2bfuoyips4rbj0vhvbp9be,-3.7016,0.4580,7.1614,0;az7owi5t4bclkamjsjuyxzv11,7.4311,23.9679,5.9402,1;6z3a27dxffwac
yxof0imvcfvu,-19.8150,11.9859,3.3955,0;axj86ms31y5wju543115j2lwq,-2.5326,26.8369,5.4973,1;9z2j13b15yh4i1vyofautc9yi,4.2968,29.
8843,6.9546,1;18qxb54ol1s4mtgl7n3224ciy,-5.6604,21.9890,4.6067,0;8u5vfczmsnklqp9bpilrbvy8k,6.6595,28.1384,6.9019,1;:8.0700,23.
7693,,4.7402,0,1,2836924213|2836924215,48gk2hpqts16p9sx9kjhaydq4,az7owi5t4bclkamjsjuyxzv11,97400:
```

The structure is also defined in and described by the XML format specification file.

Frame Count

Unique numeric identifier (incremental counter) for data frame (integer). For example, 15819. The first frame of the 1st half is always 10000, of the 2nd 1000000.

Player Tracking Data

The section covers data for all players present on the field in a given frame. It consists of semicolon-separated values, where each value describes tracking data for a single player and is internally structured to comma-separated values:

[Player UUID],[Player X coordinate],[Player Y coordinate],[Player speed],[Player visible in broadcast];[next player data];...

- Player UUID is a SDAPI player UUID (string).
- Player X,Y coordinates are real-world coordinates in meters in cartesian coordinate system with origin [0,0] in the center of the pitch (double).
- Player speed is in meters per second (double).
- Player visible in broadcast value is `1` if the player was detected from broadcast or `0` if the player was added by imputation model (boolean).

For example, for one player: `ceervf28fygp7m2euv5n5uu39,7.8188,18.8953,5.1791,1;`

Ball Tracking And Frame-Level Data

The section consists of comma-separated values:

[Ball X coordinate],[Ball Y coordinate],[Ball Z coordinate],[Ball speed],
[Ball visible in broadcast],[Match period],[Opta events],[Team in possession],[Player in possession],
[Video time ms]

- Ball X,Y coordinates are real-world coordinates in meters in cartesian coordinate system with origin [0,0] in the center of the pitch (double).

- Ball Z coordinate is always empty, because the value is currently not available (double).
- Player speed is in meters per second (double).
- Ball visible in broadcast value is always 0, because ball is always added by imputation model (boolean).
- Match period is numeric match period value - 1 - first half, 2 - second half (integer).
- Opta events is a list of opta event identifiers corresponding to the tracking frame, delimited by |, or empty when no opta event is available (string).
- Team in possession, UUID or empty (string).
- Player in possession, UUID or empty (string).
- Video time ms is a value representing the match clock including milliseconds as unix timestamp units, which can be used to determine exact point in broadcast video (integer).

For example, 8.0700,23.7693,,4.7402,0,1,2836924213|2836924215,48gk2hpqts16p9sx9kjhaydq4,az7owi5t4bclkamjsjuyxzv1l,97400

History Of Data Format Versions

Data format version is specified in metadata file, in `<Metadata>` → `<GlobalConfig>` → `<Version>`. Possible values:

- **2.2.0** : Current version. Format is the same as **2.0.2**, but the newest imputation model is used. Games from season 2024 onwards use this version.
- **2.0.2** : Added player and ball speed. Only some games from season 2023 use this version.
- **2.0.1**, **2.0.0**, **1.0.0** : Unused anymore.